- **Step 2. Parameter estimation:** Estimate candidate models and examine coefficients. The goal is to find a stationary and parsimonious model with a good fit.
- **Step 3. Diagnostic checking:** Assess model adequacy by checking whether residuals resemble white noise. If necessary, return to Step 1.

- Once residuals look like white noise, one may proceed to forecasting (Step 4).

8.1 Model Selection

Generally speaking, two approaches are available for determining the orders of ARMA models.

8.1.1 ACF/PACF

- One may select the lag orders by comparing the sample ACF/PACF with theoretical ACF/PACF.

- ACF is useful in selecting the order of an MA model, whereas PACF is useful in selecting the order of an AR model.

- The *partial correlation* in general is a conditional correlation—the correlation between two variables while taking into account the values of the some other variables.

- For example, in the context of a regression model, $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon_t$, the partial correlation between y and x_3 is the correlation between y and x_3 taking into account how both of them are related to x_1 and x_2:

$$\frac{Cov\left(y, x_3 | x_1, x_2\right)}{\sqrt{Var\left(y|x_1, x_2\right) Var\left(x_3|x_1, x_2\right)}},$$

  which can be computed by the correlation between two residuals: residuals from the regression of y on x_1 and x_2, and residuals from the regression of x_3 on x_1 and x_2.

- For a time series $\{y_t\}$, the *Partial Autocorrelation Function* (PACF) between y_t and y_{t-h} is defined as the correlation between y_t and y_{t-h} conditional on all observations between the time t and $t-h$, namely, $y_{t-h+1}, y_{t-h+2}, \ldots, y_{t-1}$. For example, the lag-1 PACF equals

to the lag-1 ACF. The lag-2 PACF is

$$\frac{Cov\left(y_t, y_{t-2}|y_{t-1}\right)}{\sqrt{Var\left(y_t|y_{t-1}\right)Var\left(y_{t-2}|y_{t-1}\right)}}.$$

The lag-3 PACF is

$$\frac{Cov\left(y_t, y_{t-3}|y_{t-1}, y_{t-2}\right)}{\sqrt{Var\left(y_t|y_{t-1}, y_{t-2}\right)Var\left(y_{t-3}|y_{t-1}, y_{t-2}\right)}},$$

and so on.

- Consider the following AR models:

$$\begin{aligned}
y_t &= \phi_{0,1} + \phi_{1,1}y_{t-1} + \varepsilon_{1t}, \\
y_t &= \phi_{0,2} + \phi_{1,2}y_{t-1} + \phi_{2,2}y_{t-2} + \varepsilon_{2t}, \\
y_t &= \phi_{0,3} + \phi_{1,3}y_{t-1} + \phi_{2,3}y_{t-2} + \phi_{3,3}y_{t-3} + \varepsilon_{3t}, \\
y_t &= \phi_{0,4} + \phi_{1,4}y_{t-1} + \phi_{2,4}y_{t-2} + \phi_{3,4}y_{t-3} + \phi_{4,4}y_{t-4} + \varepsilon_{4t}, \\
&\vdots
\end{aligned}$$

The estimate $\hat{\phi}_{1,1}$ in the first equation is the lag-1 sample PACF of y_t. The estimate $\hat{\phi}_{2,2}$ in the second equation is the lag-2 sample PACF of y_t. The estimate $\hat{\phi}_{3,3}$ in the third equation is the lag-3 sample PACF of y_t, and so on.

- For an $AR\left(p\right)$ model, the lag-h PACF should be zero for all $h > p$. (The sample PACF will not be zero, but will be close to zero for $h > p$.)

- Summary

Table 1: ACF and PACF of ARMA models

Model	ACF	PACF
$MA\left(q\right)$	cuts off after lag q	tails off
$AR\left(p\right)$	tails off	cuts off after lag p
$ARMA\left(p,q\right)$	tails off	tails off

8.1.2 Information criteria

- Mixed ARMA models are often difficult to identify using the ACF/PACF.

- A number of information criteria have been proposed. Among them two popular choices are the *Akaike Information Criterion* (AIC) and *Schwarz-Bayesian Information Criterion* (BIC). Models are selected by minimizing the AIC/BIC.

- The AIC is defined as

$$AIC = -2\log L + 2k,$$

  where L is the likelihood function evaluated at the maximum likelihood estimates, k is the number of free parameters (including σ^2, and intercept if it is present) . The first term measures the goodness of fit, whereas the second term, known as the *penalty function* of the criterion, penalizes a candidate model by the number of parameters used.

- The BIC is defined as

$$BIC = -2\log L + k\log T = AIC + (\log T - 2)\, k.$$

- BIC imposes a higher penalty on model complexity and tends to select lower lag orders than AIC when T is moderate or large.

- AIC is an estimate of the divergence between the fitted model and the true conditional density. A lower value of AIC means that a model is considered to be closer to the truth. By contrast, BIC is an estimate of a function of the posterior probability of a model being true. A lower value of BIC means that a candidate model is deemed to be more likely to be the true model.

- BIC is consistent in the sense that the probability of selecting the true model orders converges to one as $T \to \infty$. In contrast, AIC is not consistent but typically has a lower probability of choosing the lag orders that are too small. Put differently, AIC has the risk of overfitting (that is, the selected model may be more complex than necessary and capture random noise in the data), whereas BIC is more prone to underfitting.

- To address potential overfitting, the corrected AIC (AICc) for ARMA models was proposed

(Hurvich and Tsai, 1989):

$$AICc = AIC + \frac{2k\,(k+1)}{T-k-1},$$

which is particularly useful when T is small, or when k is a moderate to large fraction of T.

- Some references/software may define AIC/BIC as $a + bAIC$ and $a + bBIC$, where a and b ($b \neq 0$) are some constants.
- A final caveat is that the sample used in estimating and comparing multiple models must be the same. That is, the same sample (the same number of observations) should be used for estimation of all models.

8.2 Diagnostic Checking

- Plot the residuals: Are there outliers? Are there periods in which the model does not fit the data well? Is the variance of residuals increasing?
- Check the ACF/PACF: Are any or all of the autocorrelations/partial autocorrelations statistically significant?
- Fit high-order models?
- Forecasting

9 Introduction to Forecasting

9.1 Loss Function and Optimal Point Forecast

- Different types of times series forecasts may be of interest:
  - *Point forecast*: A single best estimate of a future value.
  - *Interval forecast*: A range of values in which a future value is expected to fall, typically with a confidence level.
  - *Density forecast*: A full probability distribution of future values.

- Suppose we are at time index t and are interested in forecasting y_{t+h}, where $h \geq 1$. The time index t is called the *forecast origin,* h the *forecast horizon,* and $t+h$ the *target date.*

- Let $\hat{y}_{t+h|t}$ denote the (conditional) forecast of y_{t+h} formed using the information available at time period t. Let Ω_t denote the information set at t, namely, $\Omega_t = (y_t, y_{t-1,} \ldots)$ (may also contain data on other variables). $\hat{y}_{t+1|t}$ is called the 1-step-ahead forecast. $\hat{y}_{t+2|t}$ is called the 2-step-ahead forecast. $\ldots$ $\hat{y}_{t+h|t}$ is called the h-step-ahead forecast.

- The h-step ahead *forecast error* is given by $e_{t+h} = y_{t+h} - \hat{y}_{t+h|t}$, that is, the difference between the observed value and its forecast. Forecast error cannot be avoided and are costly.

- What would be an "optimal" forecast? It depends on the *loss function*, denoted by $L(e)$, which measures the loss (cost) due to potential forecast error.

- The most commonly used loss function is the *quadratic loss function*, which is defined as

$$L(e_{t+h}) = L\left(y_{t+h}, \hat{y}_{t+h|t}\right) = e_{t+h}^2 = \left(y_{t+h} - \hat{y}_{t+h|t}\right)^2.$$

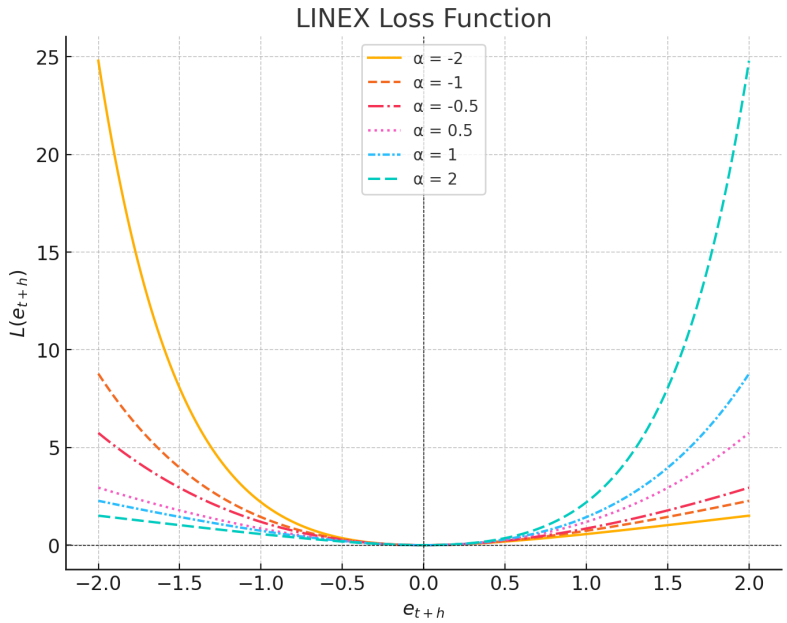
  - Quadratic loss is symmetric (treating positive and negative forecast errors symmetrically).
  - Quadratic loss penalizes large errors much more than small errors.

- Another well-known symmetric loss function is the *absolute loss* function, defined as $L(e_{t+h}) = |e_{t+h}|$, where the loss increases at a constant (linear) rate with the magnitude of the error.

- An important example of an asymmetric loss function is the *linear exponential (LINEX)* function:

$$L(e_{t+h}; \alpha) = \frac{2\left[\exp(\alpha e_{t+1}) - \alpha e_{t+1} - 1\right]}{\alpha^2},$$

  where α is a parameter that controls the degree of asymmetry.



- If $\alpha > 0$, large under-predictions incur a higher cost than over-predictions of the same magnitude. The reverse holds when $\alpha < 0$. Note that $\lim_{\alpha \to 0} L\left(e_{t+h}; \alpha\right) = e_{t+1}^2$.

- The *optimal forecast* of y_{t+h} is obtained by minimizing the expected loss conditional on the information available at t:

$$\hat{y}_{t+h|t}^{*} = \underset{\hat{y}_{t+h|t}}{\operatorname{argmin}} \left\{ E\left[L\left(y_{t+h}, \hat{y}_{t+h|t} \right) | \Omega_t \right] \right\}.$$

- Under quadratic loss, $E\left(y_{t+h}|\Omega_t\right)$ is the optimal forecast of y_{t+h} conditional on Ω_t, namely, $\hat{y}_{t+h|t}^{*} = E\left(y_{t+h}|\Omega_t\right)$, which is said to be optimal in the mean squared error (MSE) sense.
  Proof.

- We will use this fundamental result to construct forecasts for ARMA processes. It should be noted that this result relies on

  – the quadratic loss function

  - the assumption that the true conditional density, $f\left(y_{t+h}|\Omega_t\right)$, is known and will not be affected by the act of forecasting

9.2 Forecast Evaluation

- In practice, the accuracy of forecasts should be evaluated on new data that were not used when fitting the model.
- A common approach is to split the available data into *training* and *test* data, where the test data are not used in determining the forecasting method and hence should reveal how well the model is likely to forecast on new data.
- The out-of-sample mean of the quadratic loss function is known as the *mean squared forecast errors* (MSFE or MSE), and its square root, *root mean squared error* (RMSE), is often reported.
- Other frequently used forecast accuracy measures include the following:
- *Mean absolute error* (MAE): mean($|e_{t+h}|$), where the mean is taken over t. Both MSE and MAE are scale-dependent, meaning they cannot be used in comparing series with different units.
- *Mean absolute percentage error* (MAPE): mean($|p_{t+h}|$), where $p_{t+h} = 100e_{t+h}/y_{t+h}$. It is unit-free, making it useful for comparing forecast performances across series with different units. However, it has limitations:
  - It is undefined if $y_{t+h} = 0$ and can take extreme values if any y_{t+h} is close to zero.
  - It is asymmetric, penalizing negative errors (when $y_{t+h} < \hat{y}_{t+h|t}$) more than on positive errors—"equal errors above the actual value result in a greater APE than those below the actual value" (Makridakis, 1993).
- *Mean absolute scaled error* (MASE): mean ($|q_j$), where the error is scaled based on the training MAE from a simple forecast method, such as the naive forecasts (Hyndman and

Koehler, 2006):

$$q_j = \frac{e_j}{\frac{1}{T-m}\sum_{t=m+1}^{T}|y_t - y_{t-m}|},$$

with m being the span of periodic seasonal behavior. $m = 1$ for non-seasonal data. A scaled error is smaller than one if it arises from a better forecast than the average naive forecast computed on the training data.

- *Root mean squared scaled error* (RMSSE): $\sqrt{\text{mean}\left(q_j^2\right)}$, where

$$q_j^2 = \frac{e_j^2}{\frac{1}{T-m}\sum_{t=m+1}^{T}(y_t - y_{t-m})^2}.$$

Both MASE and RMSSE are unit-free.

9.3 Forecasting with AR Models

9.3.1 Forecasting with AR(1)

- Consider the stationary $AR(1)$ model: $y_t = \phi_0 + \phi_1 y_{t-1} + \varepsilon_t$, $|\phi_1| < 1$. Updating one period yields

$$y_{t+1} = \phi_0 + \phi_1 y_t + \varepsilon_{t+1}.$$

Then

$$\begin{aligned}\hat{y}_{t+1|t} &= E(y_{t+1}|\Omega_t) \\ &= \phi_0 + \phi_1 E(y_t|\Omega_t) + E(\varepsilon_{t+1}|\Omega_t) \\ &= \phi_0 + \phi_1 y_t.\end{aligned}$$

Similarly,

$$\begin{aligned}\hat{y}_{t+2|t} &= E\left(y_{t+2}|\Omega_t\right) \\ &= \phi_0 + \phi_1 E\left(y_{t+1}|\Omega_t\right) + E\left(\varepsilon_{t+2}|\Omega_t\right) \\ &= \phi_0 + \phi_1 \hat{y}_{t+1} \\ &= \phi_0 + \phi_1 E\left(\phi_0 + \phi_1 y_t|\Omega_t\right) \\ &= \phi_0\left(1+\phi_1\right) + \phi_1^2 y_t.\end{aligned}$$

By forward iteration,

$$\begin{aligned}\hat{y}_{t+h|t} &= E\left(y_{t+h}|\Omega_t\right) = \phi_0 + \phi_1 \hat{y}_{t+h-1|t} \\ &= \phi_0\left(\frac{1-\phi_1^h}{1-\phi_1}\right) + \phi_1^h y_t. \end{aligned} \tag{22}$$

Since $|\phi_1| < 1$, $E\left(y_{t+h}|\Omega_t\right)$ converges to $\mu = \frac{\phi_0}{1-\phi_1}$, as $h \to \infty$. Therefore, a stationary process is known as a *mean reverting* process, meaning the long-term point forecast, $\lim_{h\to\infty} E\left(y_{t+h}|\Omega_t\right)$, approaches its unconditional mean.

- Let $x_t = y_t - \mu$ be the mean-adjusted series. Then $\hat{x}_{t+h|t} = \phi_1^h x_t$. The speed of mean reversion is measured by the *half-life*, which is the forecast horizon such that $\hat{x}_{t+h|t} = \frac{1}{2}x_t$. That is, $\phi_1^h = \frac{1}{2}$, and hence $h = \ln(0.5) / \ln(\phi_1)$.

- The quality of the forecasts deteriorates as we forecast further into the future. To see this, the 1-step-ahead forecast error is

$$e_{t+1} = y_{t+1} - \hat{y}_{t+1|t} = \left(\phi_0 + \phi_1 y_t + \varepsilon_{t+1}\right) - \left(\phi_0 + \phi_1 y_t\right) = \varepsilon_{t+1}.$$

The 2-step-ahead forecast error is given by

$$\begin{aligned} e_{t+2} &= y_{t+2} - \hat{y}_{t+2|t} = (\phi_0 + \phi_1 y_{t+1} + \varepsilon_{t+2}) - \left(\phi_0 + \phi_1 \hat{y}_{t+1|t}\right) \\ &= \varepsilon_{t+2} + \phi_1 \left(y_{t+1} - \hat{y}_{t+1|t}\right) \\ &= \varepsilon_{t+2} + \phi_1 \varepsilon_{t+1}. \end{aligned}$$

The 3-step-ahead forecast error is given by

$$\begin{aligned} e_{t+3} = y_{t+3} - \hat{y}_{t+3|t} &= (\phi_0 + \phi_1 y_{t+2} + \varepsilon_{t+3}) - \left(\phi_0 + \phi_1 \hat{y}_{t+2|t}\right) = \varepsilon_{t+3} + \phi_1 e_{t+2}. \\ &= \varepsilon_{t+3} + \phi_1 \left(y_{t+2} - \hat{y}_{t+2|t}\right) \\ &= \varepsilon_{t+3} + \phi_1 e_{t+2} = \varepsilon_{t+3} + \phi_1 \varepsilon_{t+2} + \phi_1^2 \varepsilon_{t+1}. \end{aligned}$$

Hence, the h-step-ahead forecast error is

$$e_{t+h} = \varepsilon_{t+h} + \phi_1 \varepsilon_{t+h-1} + \phi_1^2 \varepsilon_{t+h-2} + \ldots + \phi_1^{h-1} \varepsilon_{t+1}. \tag{23}$$

It can be seen from (23) that $E\left(e_{t+h}|\Omega_t\right) = 0$ and hence $\hat{y}_{t+h|t}$ is unbiased for $y_{t+h|t}$, for all $h \geq 1$.

- The variances of the forecast errors are

$$\begin{aligned} Var\left(e_{t+1}\right) &= \sigma^2, \\ Var\left(e_{t+2}\right) &= \sigma^2 \left(1 + \phi_1^2\right) \geq Var\left(e_{t+1}\right), \\ Var\left(e_{t+3}\right) &= \sigma^2 \left(1 + \phi_1^2 + \phi_1^4\right) \geq Var\left(e_{t+2}\right), \\ &\vdots \\ Var\left(e_{t+h}\right) &= \sigma^2 \left(1 + \phi_1^2 + \phi_1^4 + \ldots + \phi_1^{2(h-1)}\right) = \sigma^2 \sum_{i=0}^{h-1} \phi_1^{2i}. \end{aligned}$$

The variance of the forecast error increases with the forecast horizon, h. As $h \to \infty$, $Var\left(e_{t+h}\right) \to \frac{\sigma^2}{1-\phi_1^2}$, which is the unconditional variance of series $\{y_t\}$.

- If $\varepsilon_t \sim N\left(0, \sigma^2\right)$, one can construct confidence intervals around the forecasts. For example, the 95% confidence interval of the h-step-ahead forecast is $\hat{y}_{t+h|t} \pm 1.96\sigma \left(\sum_{i=0}^{h-1} \phi_1^{2i}\right)^{1/2}$.

- Replacing the unknown parameters, ϕ_0 and ϕ_1, in (22) with their estimates (by YW/MLE/CMLE etc.) gives the (feasible) h-step-ahead forecast.

- If y_t contains a deterministic (perfectly predictable) component, d_t, and follows $y_t - d_t = \phi_1 \left(y_{t-1} - d_{t-1}\right) + \varepsilon_t$, then

$$\hat{y}_{t+h|t} = E\left(y_{t+h}|\Omega_t\right) = d_{t+h} + \phi_1^h \left(y_t - d_t\right),$$

  which converges to the its perfectly predictable component as $h \to \infty$, if $|\phi_1| < 1$. By contrast, if $\phi = 1$, $E\left(y_{t+h}|\Omega_t\right) = d_{t+h} + \left(y_t - d_t\right)$ is no longer mean reverting.

9.3.2 Forecasting with AR(p)

- For higher-order models, the h-step-ahead forecasts can be obtained recursively similarly as for $AR\left(1\right)$.

- For example, for an $AR\left(2\right)$ process, $y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$,

$$\begin{aligned}
\hat{y}_{t+1|t} &= \phi_0 + \phi_1 y_t + \phi_2 y_{t-1}, \\
\hat{y}_{t+2|t} &= \phi_0 + \phi_1 \hat{y}_{t+1|t} + \phi_2 y_t, \\
\hat{y}_{t+h|t} &= \phi_0 + \phi_1 \hat{y}_{t+h-1|t} + \phi_2 \hat{y}_{t+h-2|t}, \text{ for } h = 3, 4, \ldots.
\end{aligned}$$

- More generally, for an $AR\left(p\right)$ process,

$$\hat{y}_{t+h|t} = \phi_0 + \sum_{i=1}^{p} \phi_i \hat{y}_{t+h-i|t}, \text{ for } h = 1, 2, \ldots,$$

  where the initial values are $\hat{y}_{t+h-i|t} = y_{t-i}$, for $h - i \leq 0$.

9.4 Forecasting with MA Models

- Consider the $MA(1)$ model, $y_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$. The 1-step-ahead forecast is

$$\hat{y}_{t+1|t} = E(y_{t+1}|\Omega_t) = \mu + E(\varepsilon_{t+1}|\Omega_t) + \theta E(\varepsilon_t|\Omega_t) = \mu + \theta\varepsilon_t,$$

  where ε_t can be computed using the AR representation of the $MA(1)$ model (assuming $|\theta| < 1$) from realizations $y_t, y_{t-1}, \ldots$. The 2-step-ahead forecast is

$$\hat{y}_{t+2|t} = E(y_{t+2}|\Omega_t) = \mu + E(\varepsilon_{t+2}|\Omega_t) + \theta E(\varepsilon_{t+1}|\Omega_t) = \mu.$$

  It is easily seen that $\hat{y}_{t+h|t} = \mu$, for $h \geq 2$. For $MA(1)$ models, mean reverting takes only one time period.

- *Exercise*: Can you show that multistep-ahead forecasts of $MA(2)$ models go to the unconditional mean after two steps?

- In general, multistep-ahead forecasts of $MA(q)$ models go to the unconditional mean after the first q steps. Intuitively, since an $MA(q)$ model has finite memory, its forecasts move towards the mean of the series quickly.

- To compute the forecasts, we need to estimate $\varepsilon_t, \varepsilon_{t-1}, \ldots$ from realizations $y_t, y_{t-1}, \ldots$. Again, this can be done if the MA model $y_t = \mu + \theta(L)\varepsilon_t$ is invertible, where $\theta(L) = \theta_0 + \theta_1 L + \theta_2 L^2 + \ldots + \theta_q L^q$, so that ε_t in practice can be computed from a truncated version of the $AR(\infty)$ representation: $\varepsilon_t = \theta(L)^{-1}(y_t - \mu)$.

- The above discussion on forecasting with pure AR and MA models can be readily generalized to forecasting using mixed $ARMA$ models.